Quiz — 1.1.1 Structure and Function of the Processor — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Answer key

Section A (8 marks — 1 each)

1. **C** — The Program Counter always holds the address of the next instruction; it is incremented each cycle.
2. **B** — The Memory Data Register holds data/instruction just fetched or about to be written. (Remember: MAR = address, MDR = data.)
3. **B** — The address bus is unidirectional, CPU → memory only.
4. **C** — After PC → MAR, a read signal is placed on the control bus and the instruction returns via the data bus into the MDR.
5. **C** — The CU decodes instructions and generates control signals to synchronise and direct the CPU.
6. **B** — Cache is small, very fast memory on/near the CPU (not part of RAM) holding frequently/recently used data.
7. **B** — More cores help only if the task is parallelisable and software supports it.
8. **B** — Pipelining overlaps fetch/decode/execute of consecutive instructions to raise throughput.

Section B (12 marks)

9. **[3]** — 1 mark each (full name + role): - MAR = **Memory Address Register** — holds the address of the location to be read from / written to. - MDR = **Memory Data Register** — holds the data/instruction just fetched, or about to be written, to memory. - CIR = **Current Instruction Register** — holds the current instruction while it is decoded/executed.

10. **[2]** — 1 mark each: Performs **arithmetic** operations (+ − × ÷); performs **logic** operations / comparisons (AND, OR, NOT, compare) and shifts; (result is placed in the ACC). Any two distinct points.

11. **[2]** — 2^{16} (1 mark for method 2^n) = **65,536 locations** (1 mark for correct answer).

12. **[2]** — 1 mark: higher clock speed = more pulses/cycles per second, so more fetch–decode–execute cycles completed per second / more instructions processed per second. 1 mark: limited by **heat / power consumption** (overheating).

13. **[2]** — 1 mark: L1 is **smaller and faster** (on the core); L3 is **larger but slower** (often shared) — though still faster than RAM. 1 mark: checked in order **L1 → L2 → L3 → RAM**.

14. **[1]** — A **new address is written into the PC** (so execution continues elsewhere); the instructions already prefetched into the pipeline must be **flushed/discarded** (branch hazard), wasting cycles.

Section C (5 marks)

15. **[5]** — Level-based / point marking, award up to 5 for a balanced, applied discussion. Indicative content: - Higher clock speed (X) means each individual core completes more FDE cycles per second — good for the **sequential / single-threaded** parts of a game (e.g. game logic, physics dependencies). - More cores (Y) allow **parallel processing** — multiple tasks (rendering prep, AI, audio, physics) run simultaneously — but **only if the game engine is written to use multiple threads**. - Larger cache (Y) reduces slow RAM fetches (fewer cache misses), improving throughput for frequently accessed data/instructions. - Trade-off: if the game is poorly multi-threaded, X's higher clock speed may win; modern game engines are increasingly parallel, favouring Y. - Justified conclusion linking the choice to the workload (e.g. "Y is likely better for a modern, multi-threaded engine because…"). Max 4 if no justified judgement/conclusion is made.

Total: 8 + 12 + 5 = 25